

Universidade de Aveiro

Mestrado em Robótica e Sistemas Inteligentes

Mobile Robotics – Assignment 1

Robot Probabilistic Localisation
using Redundant Landmarks

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Abstract

This report describes the implementation of a probabilistic localisation system for a mobile robot navigating through a set of fixed beacons. A smooth trajectory is planned from the origin $(0,0,0)$ through all N beacon positions using Piecewise Cubic Hermite interpolation (`pchip`). An Extended Kalman Filter (EKF) fuses the unicycle motion model with range and bearing measurements from all visible beacons at each time step, gracefully ignoring any beacon that returns `NaN`. Running with $N=12$ beacons the filter achieves a mean position error of 0.96 m and a mean heading error of 0.009 rad over a path of ≈ 851 m. Wheel velocities are also computed for three kinematic models – Differential Drive, Tricycle and Omnidirectional – and saved to the required output files.

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Chapter 1. Introduction

The goal of this assignment is to localise a mobile robot that travels along a path which starts at a fixed reference point and visits N beacons (landmarks) of known position before stopping at the last one. Beacons can be sensed by an on-board sensor that returns the range d_i and the bearing α_i to each beacon i , both corrupted by Gaussian noise. Because more beacons than strictly necessary are typically visible at any instant, the system can exploit this *redundancy* to produce a more robust estimate. Measurements may also be temporarily unavailable (returned as NaN) when a beacon is too close, too far, or fails intermittently.

Two broad localisation strategies are contrasted in this work:

- **Classic deterministic methods** (triangulation, trilateration) – produce a single point estimate from a minimal set of measurements with no uncertainty quantification.
- **Probabilistic EKF** – maintains a full Gaussian belief over the pose $\mathbf{x}_k = (x_k, y_k, \theta_k)^\top$, propagated by a nonlinear motion model and corrected at every sensor update using all available beacon measurements simultaneously.

The report is structured as follows. Chapter 2 presents the mathematical foundations of trajectory planning, the EKF formulation, and the three kinematic models. Chapter 3 summarises the numerical results and discusses their quality. Chapter 4 draws conclusions and suggests directions for future work.

Chapter 2. Methodology

2.1 Trajectory Planning

2.1.1 Requirements

The planned trajectory must:

1. start at the origin with pose $(0, 0, 0)$;
2. pass through every beacon position $B_n = (X_n, Y_n)$, $n = 1, \dots, N$, in order;
3. end at the last beacon;
4. be smooth, using *Hermite cubic interpolation* (**pchip**) as the required interpolation method;
5. contain control points evenly spaced in x , with y values obtained from the **pchip** curve.

2.1.2 Control-point spacing

Given a desired average linear velocity V (m/s) and sensor sampling interval Δt (s), the nominal arc-length between successive control points is

$$\Delta d = V \cdot \Delta t \quad [\text{m}]. \quad (2.1)$$

For each segment between consecutive waypoints the number of intermediate control points is approximately $\lfloor \ell / \Delta d \rfloor$, where ℓ is the Euclidean distance between the waypoints.

2.1.3 Hermite interpolation

The full ordered waypoint list is

$$\mathbf{W} = \begin{pmatrix} 0 & 0 \\ X_1 & Y_1 \\ \vdots & \vdots \\ X_N & Y_N \end{pmatrix} \in \mathbb{R}^{(N+1) \times 2}. \quad (2.2)$$

A uniformly spaced query vector $x_q \in [\min x_w, \max x_w]$ is constructed and the corresponding y values are

$$y_q = \text{pchip}(x_w, y_w, x_q), \quad (2.3)$$

yielding M control points $(x_q^{(i)}, y_q^{(i)})$. The heading at each point is set to the direction to the next point:

$$\theta_i = \text{atan2}(y_q^{(i+1)} - y_q^{(i)}, x_q^{(i+1)} - x_q^{(i)}), \quad i = 1, \dots, M-1, \quad (2.4)$$

with $\theta_M = \theta_{M-1}$, and all angles wrapped to $(-\pi, \pi]$.

2.2 Extended Kalman Filter

2.2.1 State and notation

The robot state is the planar pose

$$\mathbf{x}_k = (x_k, y_k, \theta_k)^\top \in \mathbb{R}^3. \quad (2.5)$$

Following the ISO 80000-2 convention used in the assignment, predicted (prior) quantities carry an overbar and updated (posterior) quantities carry a hat.

2.2.2 Initialisation

$$\hat{\mathbf{x}}_0 = \mathbf{0}, \quad \mathbf{P}_0 = \text{diag}(\sigma_{x,0}^2, \sigma_{y,0}^2, \sigma_{\theta,0}^2) = \text{diag}(0.01^2, 0.01^2, 0.1^2). \quad (2.6)$$

2.2.3 Prediction step

The unicycle (differential-drive) motion model propagates the state by

$$\bar{x}_{k+1} = \hat{x}_k + v \Delta t \cos \hat{\theta}_k, \quad (2.7)$$

$$\bar{y}_{k+1} = \hat{y}_k + v \Delta t \sin \hat{\theta}_k, \quad (2.8)$$

$$\bar{\theta}_{k+1} = \hat{\theta}_k + \omega \Delta t, \quad (2.9)$$

where (v, ω) are the linear and angular velocity commands. The control input $\mathbf{u}_k = (v, \omega)^\top$ is derived from consecutive poses on the planned trajectory.

The covariance is propagated via linearisation:

$$\bar{\mathbf{P}}_{k+1} = \mathbf{F}_x \hat{\mathbf{P}}_k \mathbf{F}_x^\top + \mathbf{F}_w \mathbf{Q} \mathbf{F}_w^\top, \quad (2.10)$$

with process-noise covariance $\mathbf{Q} = \text{diag}(\sigma_v^2, \sigma_\omega^2)$ and Jacobians

$$\mathbf{F}_x = \begin{pmatrix} 1 & 0 & -v \Delta t \sin \hat{\theta}_k \\ 0 & 1 & v \Delta t \cos \hat{\theta}_k \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{F}_w = \begin{pmatrix} \Delta t \cos \hat{\theta}_k & 0 \\ \Delta t \sin \hat{\theta}_k & 0 \\ 0 & \Delta t \end{pmatrix}. \quad (2.11)$$

2.2.4 Update step

For each beacon i with a valid measurement (i.e. $B_i.d \neq \text{NaN}$ and $B_i.a \neq \text{NaN}$), the predicted observation from the prior state $\bar{\mathbf{x}}_{k+1}$ is

$$\delta x_i = X_i - \bar{x}_{k+1}, \quad \delta y_i = Y_i - \bar{y}_{k+1}, \quad (2.12)$$

$$\hat{d}_i = \sqrt{\delta x_i^2 + \delta y_i^2}, \quad (2.13)$$

$$\hat{\alpha}_i = \text{atan2}(\delta y_i, \delta x_i) - \bar{\theta}_{k+1}. \quad (2.14)$$

The innovation for beacon i is

$$\boldsymbol{\nu}_i = \begin{pmatrix} d_i^{\text{meas}} - \hat{d}_i \\ \text{wrap}(\alpha_i^{\text{meas}} - \hat{\alpha}_i) \end{pmatrix}, \quad (2.15)$$

where $\text{wrap}(\cdot)$ maps angles to $(-\pi, \pi]$. All visible beacons are stacked into a single innovation vector $\boldsymbol{\nu}$ and the corresponding Jacobian matrix

$$\mathbf{H}_i = \begin{pmatrix} -\delta x_i / \hat{d}_i & -\delta y_i / \hat{d}_i & 0 \\ \delta y_i / \hat{d}_i^2 & -\delta x_i / \hat{d}_i^2 & -1 \end{pmatrix} \quad (2.16)$$

is assembled into the full Jacobian $\mathbf{H}$ by row-stacking. The per-beacon noise covariance $\mathbf{R}_i = \text{diag}(\sigma_{d,i}^2, \sigma_{\alpha,i}^2)$ comes directly from the `BeaconDetection` function fields `B.dn` and `B.an`.

The standard EKF correction then gives

$$\mathbf{S} = \mathbf{H} \bar{\mathbf{P}}_{k+1} \mathbf{H}^\top + \mathbf{R}, \quad (2.17)$$

$$\mathbf{K} = \bar{\mathbf{P}}_{k+1} \mathbf{H}^\top \mathbf{S}^{-1}, \quad (2.18)$$

$$\hat{\mathbf{x}}_{k+1} = \bar{\mathbf{x}}_{k+1} + \mathbf{K} \boldsymbol{\nu}, \quad (2.19)$$

$$\hat{\mathbf{P}}_{k+1} = (\mathbf{I} - \mathbf{K} \mathbf{H}) \bar{\mathbf{P}}_{k+1} (\mathbf{I} - \mathbf{K} \mathbf{H})^\top + \mathbf{K} \mathbf{R} \mathbf{K}^\top. \quad (2.20)$$

Equation (2.20) uses the numerically stable *Joseph form*, which guarantees positive semi-definiteness even with finite-precision arithmetic. If no beacon is visible at a given step, the posterior is set equal to the prior: $\hat{\mathbf{x}}_{k+1} = \bar{\mathbf{x}}_{k+1}$, $\hat{\mathbf{P}}_{k+1} = \bar{\mathbf{P}}_{k+1}$.

2.2.5 Use of estimated poses as control input

In a real deployment the ground-truth pose is unavailable. Control inputs (v, ω) are therefore derived from the *current estimate* $\hat{\mathbf{x}}_k$ and the desired next waypoint $\mathbf{x}_{k+1}^{\text{plan}}$:

$$v = \frac{\|\mathbf{x}_{k+1}^{\text{plan}} - \hat{\mathbf{x}}_k\|_2}{\Delta t_{\text{actual}}}, \quad \omega = \frac{\text{wrap}(\theta_{k+1}^{\text{plan}} - \hat{\theta}_k)}{\Delta t_{\text{actual}}}, \quad (2.21)$$

where $\Delta t_{\text{actual}} = \|\mathbf{p}_{k+1}^{\text{plan}} - \hat{\mathbf{p}}_k\|_2/V$ and $\mathbf{p}$ denotes the (x, y) part.

2.3 Kinematic Models

All three robots share parameters: wheel radius r and inter-wheel distance L . At each trajectory step i , the linear velocity v_i and angular velocity ω_i are extracted from consecutive ground-truth poses, and the corresponding wheel quantities are computed as described below.

2.3.1 Differential Drive

The inverse kinematics from (v, ω) to right/left wheel angular velocities is

$$\omega_R = \frac{v + \omega L/2}{r}, \quad \omega_L = \frac{v - \omega L/2}{r}. \quad (2.22)$$

Verification: for pure straight motion ($\omega = 0$) both wheels spin at v/r ; for pure in-place rotation ($v = 0$) the wheels spin at equal speed in opposite directions.

2.3.2 Tricycle

The tricycle model assumes rear-wheel traction on the equivalent single virtual rear wheel and a steerable front wheel. The inverse kinematics are

$$\omega_T = \frac{v}{r}, \quad \alpha = \text{atan2}(\omega L, v). \quad (2.23)$$

For pure straight motion $\alpha = 0$; for pure rotation the front wheel turns to $\pm\pi/2$.

2.3.3 Omnidirectional (3-wheel)

Using the configuration from `localvels.m` (case 3), the forward-kinematics matrix relating wheel angular velocities to the robot-frame velocity $(V_x, V_y, \omega)^\top$ is

$$\mathbf{M} = \begin{pmatrix} 0 & r/\sqrt{3} & -r/\sqrt{3} \\ -2r/3 & r/3 & r/3 \\ r/(3L) & r/(3L) & r/(3L) \end{pmatrix}. \quad (2.24)$$

In the robot frame the commanded velocity is $(v, 0, \omega)^\top$ (pure forward motion in X_r with rotation ω), so the required wheel velocities are

$$\begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \mathbf{M}^{-1} \begin{pmatrix} v \\ 0 \\ \omega \end{pmatrix}. \quad (2.25)$$

2.3.4 Last-point convention

At the final trajectory point the robot has stopped, so all wheel velocities and steering angles are set to zero.

Chapter 3. Results and Discussion

3.1 Simulation parameters

Table 3.1 summarises the default parameters used for the main run.

Table 3.1: Simulation parameters for `rm1_129466(12)`.

Parameter	Symbol	Value
Number of beacons	N	12
Sensor sampling interval	Δt	1 s
Wheel radius	r	0.15 m
Wheel separation	L	1 m
Velocity noise (1σ)	σ_v	0.1 m/s
Angular-vel. noise	σ_ω	0.1 rad/s
Desired linear velocity	V	5 m/s

3.2 Trajectory

The planner generated $M = 84$ control points over a total path length of approximately 851.6 m, starting at (0,0) and ending at (292.16, 53.16) m after visiting all 12 beacons. The spacing $\Delta d = V \cdot \Delta t = 5$ m between consecutive points approximates constant-velocity sampling.

3.3 Localisation accuracy

Table 3.2 reports the localisation errors computed against the ground-truth trajectory.

Table 3.2: EKF localisation error statistics ($N = 12$, 84 points).

Metric	Mean	Maximum
Position error $\ \hat{\mathbf{p}}_k - \mathbf{p}_k\ $	0.96 m	3.53 m
Heading error $ \hat{\theta}_k - \theta_k $	0.009 rad	0.024 rad

The position error remains below 4 m throughout a path of more than 850 m, which corresponds to a relative error under 0.5%. The peak occurs at trajectory point 83–84, where the path makes a sharp turn near the last beacon and the visible beacon geometry momentarily degrades – an expected behaviour with any filter relying on angular resection. The heading is estimated with sub-degree accuracy at all times, confirming that the bearing measurements from multiple beacons strongly constrain θ .

The EKF covariance $\hat{\mathbf{P}}_k$ correctly reflects these observations: it decreases rapidly during the first few steps as measurements arrive, and occasionally grows slightly during segments where all beacons are at the boundary of their detection range.

3.4 Kinematic model analysis

3.4.1 Differential Drive

Using Eq. (2.22), the wheel velocities vary significantly along the path. The first step requires very high $\omega_R = \omega_L \approx 537$ rad/s because the robot moves almost vertically ($\theta_1 \approx 1.53$ rad) from the origin, corresponding to a large displacement in one time step. Along straight segments both wheels spin at identical speed; at turns they diverge by an amount proportional to the curvature.

3.4.2 Tricycle

The rear-wheel speed $\omega_T = v/r$ mirrors the linear velocity profile. The steering angle α is near zero on straight segments and reaches $\pm\pi/2$ during tight turns. The `atan2` formulation ensures α is always well defined, including for pure rotation ($v = 0$) where it saturates at $\pm\pi/2$.

3.4.3 Omnidirectional

The omnidirectional drive can combine translation and rotation independently. As the trajectory has $V_y = 0$ in the robot frame, wheels 2 and 3 carry symmetric forward-thrust components while wheel 1 absorbs the purely rotational contribution. The large first-step velocities observed in the Differential Drive appear here as well, since they reflect the same large v and ω commands.

3.5 Comparison with classic methods

3.5.1 Triangulation

Triangulation recovers the robot position by intersecting bearing lines from at least two beacons. For each beacon pair (i, j) the pose (x, y) satisfying

$$\alpha_i^{\text{meas}} = \text{atan2}(Y_i - y, X_i - x) - \theta, \quad (3.1)$$

$$\alpha_j^{\text{meas}} = \text{atan2}(Y_j - y, X_j - x) - \theta, \quad (3.2)$$

must be solved jointly with an assumed θ . This is sensitive to bearing noise and geometry (small intersection angle $\Rightarrow$ large positional uncertainty); it yields no covariance estimate; and it requires a minimum of two beacons simultaneously visible.

3.5.2 Trilateration

Trilateration uses range circles: for at least three beacons

$$(x - X_i)^2 + (y - Y_i)^2 = d_i^2, \quad i = 1, 2, 3. \quad (3.3)$$

Subtracting pairs of equations linearises the problem but amplifies measurement noise when beacons are nearly collinear (poor GDOP – Geometric Dilution Of Precision). Again, no uncertainty is propagated and θ is not estimated.

3.5.3 Advantages of the EKF

1. **Uncertainty quantification.** The posterior covariance $\hat{\mathbf{P}}_k$ provides a confidence region around the estimate, enabling safety-aware planning.
2. **Motion-model integration.** The prediction step propagates the state between sensor updates, so the estimate remains useful even when all beacons are temporarily lost.
3. **Redundancy exploitation.** All N_{vis} visible beacons contribute simultaneously to the update, improving accuracy and robustness beyond what any two- or three-beacon deterministic approach can achieve.
4. **NaN robustness.** Unavailable measurements are cleanly excluded from $\mathbf{H}$ and $\boldsymbol{\nu}$; the filter degrades gracefully to a pure-prediction mode.
5. **Full pose estimation.** The EKF jointly estimates (x, y, θ) , whereas trilateration estimates only (x, y) and triangulation requires prior knowledge of θ .

The main limitations of the EKF are: (i) it assumes Gaussian noise – non-Gaussian tails can degrade performance; (ii) linearisation errors accumulate for highly nonlinear trajectories; (iii) it requires a reasonably accurate initial pose. Alternatives such as Unscented Kalman Filters (UKF) or Particle Filters (PF) can address these at higher computational cost.

Chapter 4. Conclusion

This work successfully implemented the three main components of a mobile-robot localisation pipeline:

1. **Trajectory planning.** A smooth `pchip`-interpolated trajectory was generated from the origin through all N beacon positions. Control points are evenly spaced in x at intervals of $\Delta d = V \cdot \Delta t$, consistent with constant-velocity sensor sampling.
2. **EKF localisation.** The filter fuses the unicycle motion model with multi-beacon range and bearing measurements. Using the numerically stable Joseph-form covariance update and robust NaN handling, it achieves mean position and heading errors of 0.96 m and 0.009 rad respectively over a 851 m path with $N = 12$ beacons.
3. **Kinematic analysis.** Wheel velocities for Differential Drive (ω_R, ω_L), Tricycle (ω_T, α), and Omnidirectional ($\omega_1, \omega_2, \omega_3$) drives were computed at every trajectory point using the appropriate inverse-kinematics relations and saved to the required comma-separated output files (`DD_129466.txt`, `TRI_129466.txt`, `OMNI_129466.txt`). The EKF results were saved to `loc_129466.txt`.

Compared to classic triangulation and trilateration, the EKF offers joint (x, y, θ) estimation, a principled treatment of measurement noise, graceful degradation when beacons are unavailable, and a formal uncertainty estimate through the covariance matrix – all of which are essential for robust real-world deployment.

Future directions include replacing the EKF with an Unscented Kalman Filter to handle stronger nonlinearities, adding a Particle Filter for multimodal uncertainty during initialisation, and integrating simultaneous localisation and mapping (SLAM) to relax the assumption of perfectly known beacon positions.